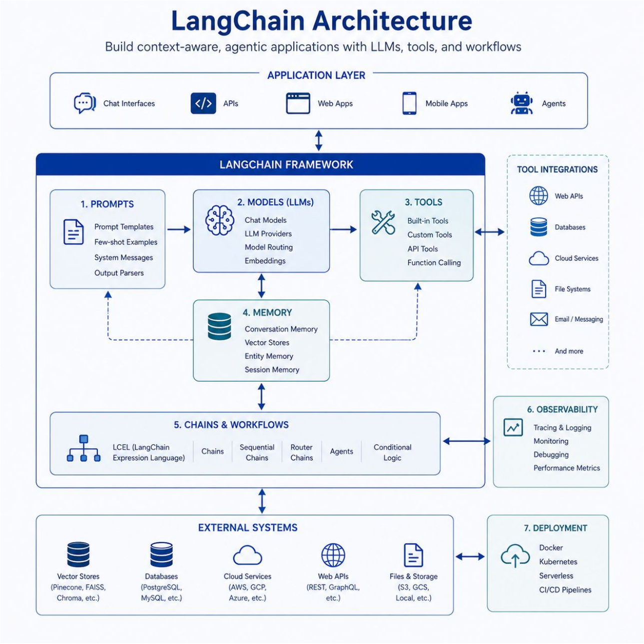
Agentic AI for Tech Professionals

Introduction to Agentic AI

What is LangChain?

LangChain is a framework for building AI-powered workflows and applications

- Simplifies LLM application development
- Connects prompts, models, and tools
- Supports chaining and orchestration
- Enables RAG and agent systems
- Helps structure AI workflows



What is LangChain? (2)

LangChain helps provide abstractions for:

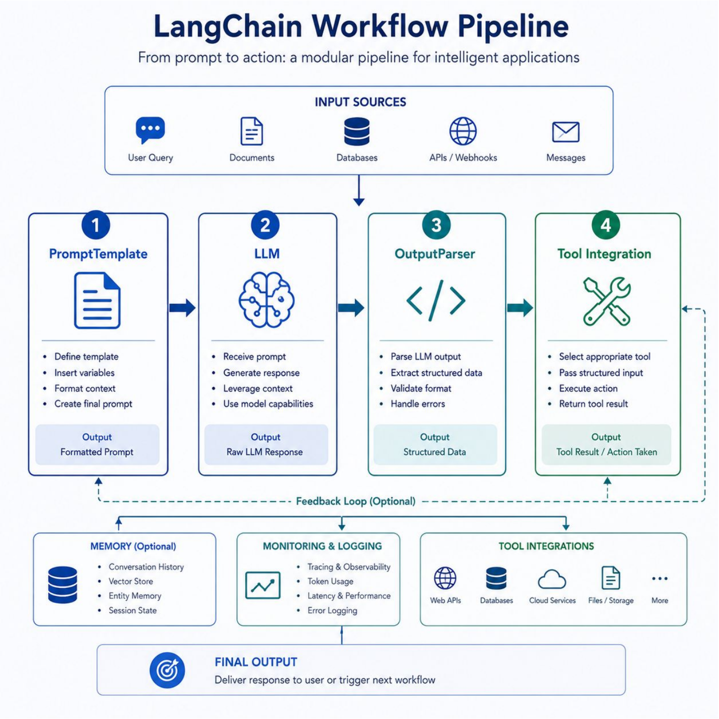
- prompts
- chains
- tools
- retrievers
- workflows

Think of LangChain as:
👉 a framework for AI workflow engineering.

How LangChain Fits into Agentic AI

LangChain helps implement the same orchestration concepts using framework abstractions

- Chains represent workflow pipelines
- Prompt templates standardize prompts
- Output parsers structure responses
- Retrievers support RAG systems
- Tools enable external integrations



How LangChain Fits into Agentic AI

LangChain does NOT replace orchestration thinking.

It implements the SAME ideas we already learned:

Input
→ Process
→ Decide
→ Act

Now using framework components:

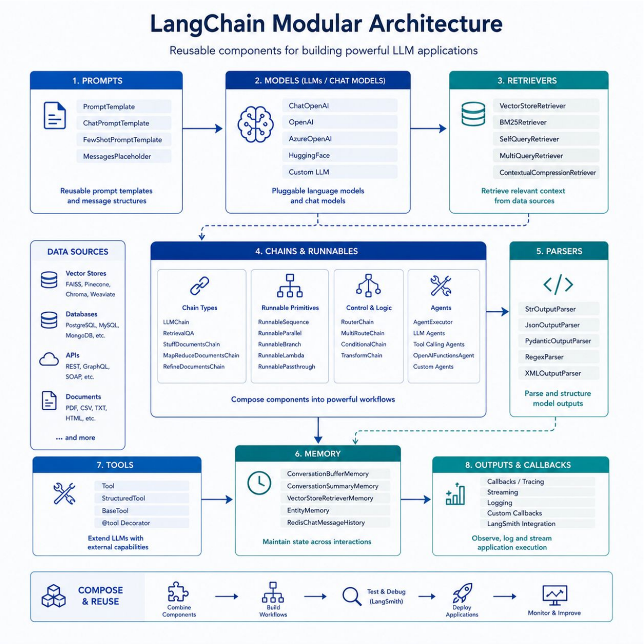
- PromptTemplate
- RunnableSequence
- OutputParser
- Retriever

This helps developers:
👉 build larger AI systems faster.

LangChain Workflow Components

LangChain applications are built from reusable workflow building blocks

- PromptTemplate creates reusable prompts
- LLM handles AI processing
- Parsers structure outputs
- Chains connect workflow stages
- Retrievers add external knowledge



LangChain Workflow Components

LangChain encourages modular design.

Example workflow:

PromptTemplate:
👉 standardized instructions

LLM:
👉 AI reasoning

OutputParser:
👉 structured JSON

Retriever:
👉 SOP knowledge retrieval

Chain:
👉 connects everything together

This is very similar to:
👉 enterprise workflow engineering.

LangChain vs Direct OpenAI API

Both approaches are useful depending on system complexity

- Direct API = simpler and lightweight
- LangChain = more structured abstraction
- Direct API easier for beginners
- LangChain useful for larger systems
- Concepts remain the same

LangChain vs Direct OpenAI API (2)

LangChain is OPTIONAL.

Many enterprise systems still use:
👉 direct API calls

Especially for:

- simple workflows
- lightweight automation
- controlled pipelines

LangChain vs Direct OpenAI API (3)

LangChain becomes useful when systems become:

- larger
- modular
- multi-step
- reusable

The key takeaway:
👉 frameworks change
👉 orchestration principles remain.

Direct OpenAI API vs LangChain Abstraction

From low-level calls to high-level orchestration

Example Use case

LangChain structures enterprise AI workflows into reusable components

- Incident input enters chain
- PromptTemplate standardizes AI logic
- AI generates structured decision
- Routing logic triggers actions
- Workflow becomes reusable and scalable

Example Use case (2)

“Machine overheating detected”

LangChain workflow:

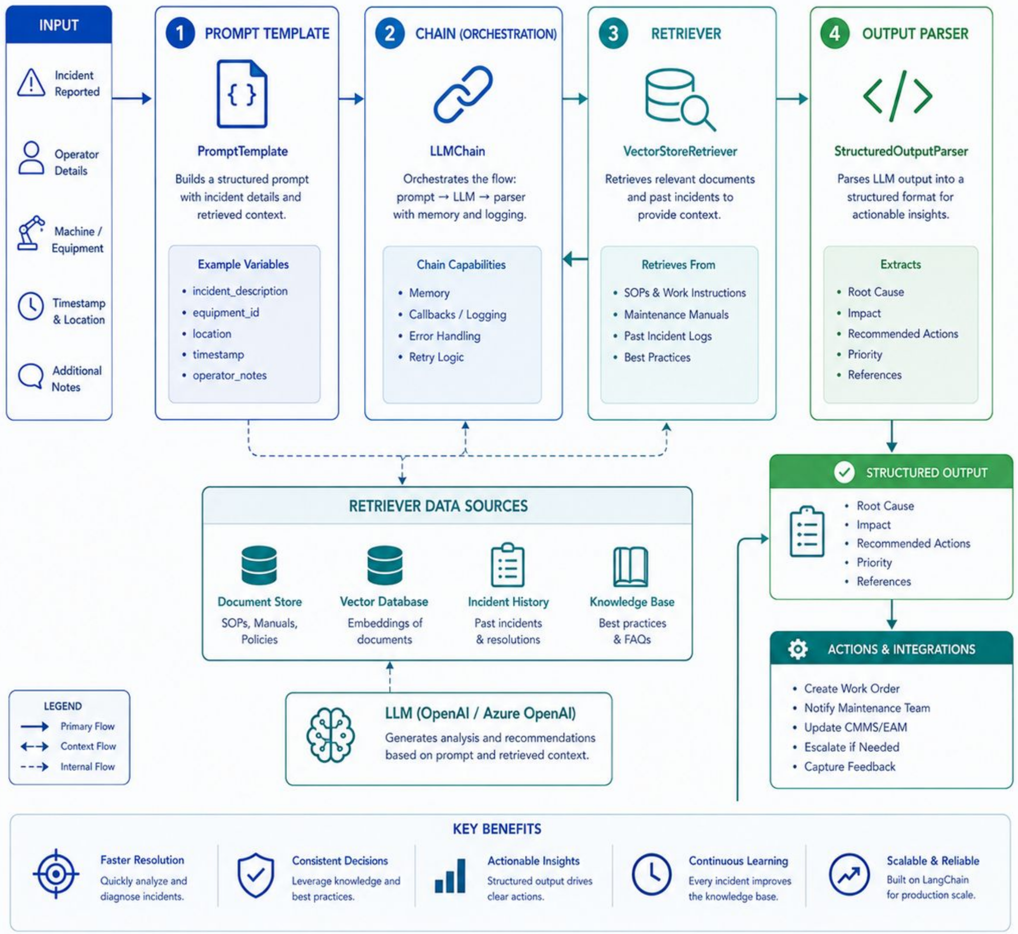
1. PromptTemplate formats request
2. LLM analyzes incident
3. Parser structures output
4. Routing logic assigns team
5. Workflow generates response

This creates:
👉 reusable enterprise AI workflows

without rewriting the entire system repeatedly.

Manufacturing Incident Workflow with LangChain

Intelligent incident analysis and resolution using LangChain components



What is LangGraph?

LangGraph extends LangChain for stateful and multi-step agent orchestration

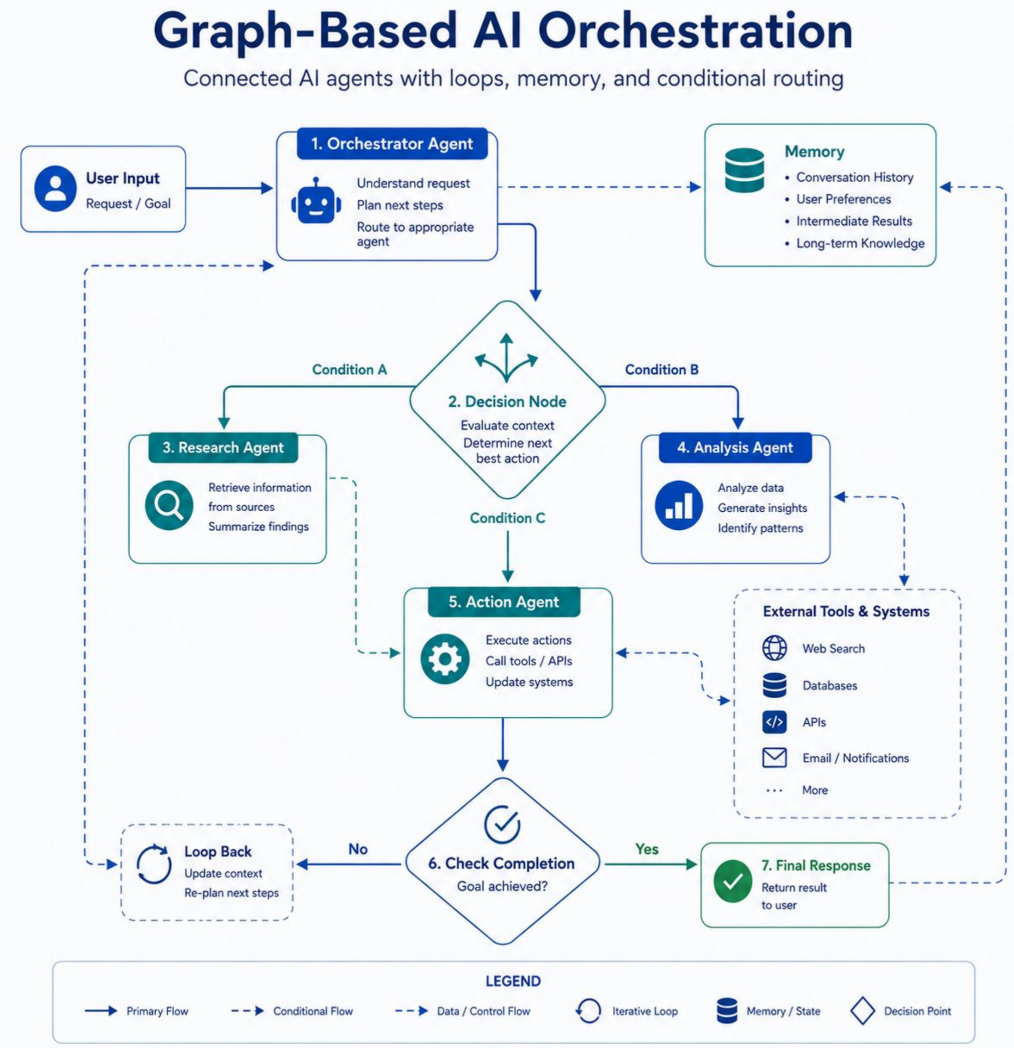
- Graph-based workflow orchestration
- Supports long-running agent flows
- Enables memory and state tracking
- Handles loops and conditional paths
- Designed for advanced agent systems

What is LangGraph? (2)

Example:
An AI agent can:

- retry actions
- revisit decisions
- maintain workflow memory
- collaborate with other agents

This is useful for:
👉 advanced enterprise AI systems.


Graph-Based AI Orchestration
Connected AI agents with loops, memory, and conditional routing
User Input
Request / Goal
1. Orchestrator Agent
Understand request
Plan next steps
Route to appropriate agent
Memory
Conversation History
User Preferences
Intermediate Results
Long-term Knowledge
Condition A
Condition B
2. Decision Node
Evaluate context
Determine next best action
3. Research Agent
Retrieve information from sources
Summarize findings
4. Analysis Agent
Analyze data
Generate insights
Identify patterns
Condition C
5. Action Agent
Execute actions
Call tools / APIs
Update systems
External Tools & Systems
Web Search
Databases
APIs
Email / Notifications
More
Loop Back
Update context
Re-plan next steps
No
6. Check Completion
Goal achieved?
Yes
7. Final Response
Return result to user
LEGEND
Primary Flow
Conditional Flow
Data / Control Flow
Iterative Loop
Memory / State
Decision Point

What is OpenClaw?

OpenClaw explores AI agents that can control software and computer interfaces

- AI agents interact with computer systems
- Can automate browser or desktop actions
- Supports tool-based execution
- Focuses on autonomous workflows
- Demonstrates future agent capabilities

What is OpenClaw?

OpenClaw represents another direction in Agentic AI:
👉 computer-use agents.

Instead of only generating text,
AI agents can:

- click interfaces
- interact with applications
- automate browser tasks
- control workflows

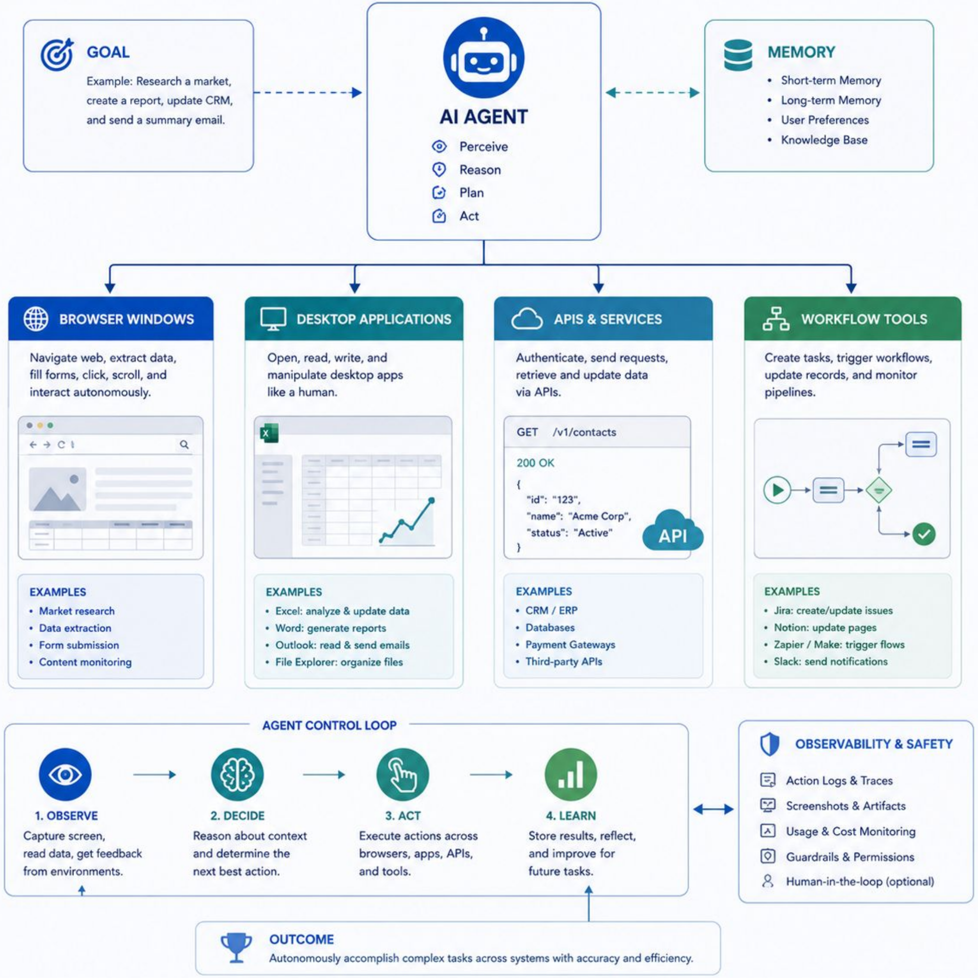
What is OpenClaw?

Example:
AI receives incident:
👉 opens maintenance system
👉 creates ticket
👉 updates dashboard
👉 sends notification

This shows where enterprise AI may evolve next:
👉 autonomous operational agents.

AI Agent: Autonomous Control Across Digital Environments

The agent perceives, decides, and acts across multiple systems to achieve goals end-to-end.



Choosing the Right Framework

Framework choice depends on workflow complexity and enterprise needs

- Direct API → simple workflows
- LangChain → modular pipelines
- LangGraph → advanced orchestration
- n8n → visual enterprise automation
- Concepts matter more than tools

Choosing the Right Framework (2)

There is no “best” framework.

Choice depends on:

- system complexity
- team skills
- operational requirements
- maintainability

Choosing the Right Framework (2)

General guideline:

👉 Direct API
Simple controlled systems

👉 LangChain
Reusable AI pipelines

👉 LangGraph
Complex multi-agent orchestration

👉 n8n
Enterprise workflow automation

Most important:
👉 understand the architecture concepts first.

Frameworks will continue changing,
but orchestration thinking remains valuable.

AI Workflow Stack: From Simple to Advanced Orchestration

Choose the right layer for your use case and scale as complexity grows

COMPLEXITY	Low	Increasing Complexity & Control		High
LAYER	1. OpenAI API	2. LangChain	3. LangGraph	4. n8n
DESCRIPTION	Direct access to OpenAI models via API. Lowest-level integration.	Framework for building LLM applications with components and chains.	Graph-based orchestration for stateful, multi-step agent workflows.	Workflow automation platform to orchestrate any tool or service.
BEST FOR	• Quick prototypes • Simple completions • Custom integrations • Maximum flexibility	• RAG applications • Prompt chaining • Tool / function calling • Memory management	• Complex agent flows • Conditional logic & loops • Multi-agent systems • Stateful workflows	• End-to-end automation • Integrations beyond LLMs • Scheduled / triggered flows • Business process automation
KEY CAPABILITIES	Model access (Chat, Completions, etc.) Fine-grained control of requests Raw API primitives	Prompt templates & chaining Tools & agents Retrievers & memory Output parsers	Graph-based workflows (nodes & edges) Loops & conditional routing State management & persistence Human-in-the-loop	Visual workflow builder 1000+ integrations Triggers & scheduling Error handling & retries
WORKFLOW STYLE	User → API Call → Response	Input → Chain → Output	Start → … → End	
WHEN TO CHOOSE	You need direct, low-level access to models with maximum control.	You're building LLM apps and need components, chains, and memory.	You need complex, stateful agent workflows with loops and conditional logic.	You want to automate end-to-end processes across many tools and services.
ABSTRACTION LEVEL	Low (You build everything)			High (More abstraction & automation)

Beyond LangChain — Modern Agent Framework Ecosystem

Enterprise AI systems can scale further using advanced orchestration frameworks

- LangGraph enables stateful agent workflows
- OpenClaw focuses on autonomous computer control
- Frameworks support advanced orchestration
- Multi-agent systems become easier to manage
- Ecosystem is evolving rapidly

LangChain Ecosystem

Building, Orchestrating, and Scaling Enterprise AI Applications